

Probability

Assignment - 2

①

$$\bar{x} = 1.8 \quad n = 50$$

$$s = 20 \text{ m} \approx 0.33 \text{ hr}$$

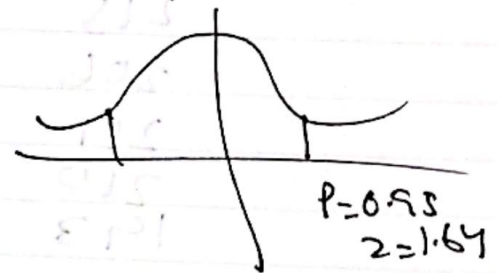
~~$$\bar{x} \pm 2 \cdot \frac{s}{\sqrt{n}}$$~~

$$\bar{x} \pm 2 \cdot \frac{s}{\sqrt{n}}$$

$$1.8 \pm 1.64 \cdot \frac{0.33}{\sqrt{50}}$$

$$\therefore 1.8 \pm 0.0765 \quad [1.7235; 1.8765]$$

Sample result contradicts the ~~report~~ report value since 2.1 hr is above the estimation.



②

$$\bar{x} = 217.7$$

$$n = 10$$

S.D

X	$(X - \bar{X})^2$
225	53.29
240	497.29
215	7.29
206	136.89
211	44.89
210	59.29
193	610.09
250	1043.29
225	53.29
202	246.49
$\Sigma = 2752.1$	

$$S = \sqrt{\frac{2752.1}{9}} = \sqrt{305.788} = 17.49$$

$$\bar{X} \pm t \cdot \frac{S}{\sqrt{n}}$$

$$= 217.7 \pm 12.51$$

$$[205.19, 230.21]$$

③

$$H_0: \mu = 42000$$

$$H_1: \mu > 42000$$

$$\bar{x} = 43260$$

$$\sigma = 5230$$

$$n = 30$$

$$z = \frac{43260 - 42000}{\frac{5230}{\sqrt{30}}}$$

$$z = 1.32$$

We fail to reject H_0 , thus alternate hypothesis is incorrect

④

$$H_0: \mu = 24$$

$$H_1: \mu > 24$$

$$\bar{x} = \frac{1445}{50} = 28.9$$

$$z = \frac{28.9 - 24}{\frac{28.7}{\sqrt{50}}} \Rightarrow 1.355$$

Excel

$$P(\bar{z} > 1.355) = 1 - Q(1.355) \\ = 0.089$$

We fail to reject H_0 since $0.089 > 0.05$.

⑤

$$H_0: \mu = 60$$

$$H_1: \mu < 60$$

$$\bar{x} = 58.875$$

$$\therefore s = \sqrt{\frac{181.35}{7}} \approx 5.09$$

$$t = \frac{58.875 - 60}{5.09/\sqrt{8}} \approx -0.625$$

$$\frac{C.V}{d.f} = 7 \quad \alpha = 0.1$$

$$t = 1.415$$

We fail to reject H_0 since

$$-0.625 > -1.415$$

Excel

⑥ Males: $\bar{x}_1 = 8.56$
 $n_1 = 50$
 $\sigma_1 = 3.3$

Females: $\bar{x}_2 = 7.94$
 $n_2 = 50$
 $\sigma_2 = 3.3$

$H_0: \mu_1 = \mu_2$

$H_1: \mu_1 > \mu_2$

$\bar{x}_1 - \bar{x}_2 = 8.56 - 7.94 = 0.64$

$z = \frac{0.62 - 0}{\sqrt{\frac{3.3^2}{50} + \frac{3.3^2}{50}}}$
 $= 0.94$

since $0.94 < 1.28$, we fail to reject H_0 .

Excel